FRONTIER RESEARCH QUESTIONS
MODULAR KOSZUL DUALITY FOR CHIRAL ALGEBRAS ON CURVES

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Contents

1. Introduction 1
2. The coderived BV/BRST identification (Resolved) 1
2.1. The problem 1
2.2. The resolution 1
2.3. Physical significance 2
3. Non-principal DS-KD commutation: the (3,2) partition 2
3.1. Context 2
3.2. What is proved 2
3.3. The open case 2
3.4. Physical motivation 2
4. The genus-5 cross-channel correction 2
4.1. The universal N-formula 2
4.2. Physical significance 3
5. Admissible Koszulness at rank ≥ 2 3
5.1. The rank obstruction 3
5.2. Physical significance 3
6. DK-4 and DK-5: Extending the Drinfeld–Kohno bridge 3
6.1. DK-4: Beyond the evaluation core 3
6.2. DK-5: Categorical E_1 primacy 3
7. Grand completion 4
8. Analytic realisation: three-layer gap 4
9. E_1 Verdier duality on ordered configurations 4
10. Resurgence: the cross-channel instanton action 4
11. Cross-channel generating function 4
12. Scalar saturation residual 5

1. Introduction

The programme of modular Koszul duality studies the bar-cobar adjunction for chiral algebras on algebraic curves, with the Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$ as the universal organising structure. The five main theorems (A–H) are proved, the shadow obstruction tower is constructed at all arities, and the genus expansion $F_g = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ is established on the uniform-weight lane. Seven major open problems were resolved in April 2026. This document records the twelve frontier questions that remain, with full mathematical context, physical motivation, and available tools.

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The standard is that of Beilinson: a claim is false until independently verified; a coincidence is an error until explained; a narrative is dangerous until every sentence is deducible from the mathematics.

2. The coderived BV/BRST identification (Resolved)

2.1. **The problem.** Let $\mathcal{A}$ be a chirally Koszul algebra on a smooth curve X, and let Σ_g denote a closed Riemann surface of genus g. The Batalin–Vilkovisky complex $\mathrm{Obs}(\Sigma_g, \mathcal{A})$ of the two-dimensional sigma model with target the chiral algebra $\mathcal{A}$ and the bar complex $B^{(g)}(\mathcal{A})$ of the genus-g factorisation homology are two chain complexes computing the same physical observables: the genus-g partition function and its deformations.

At genus 0, the identification $\mathrm{Obs}(\Sigma_0, \mathcal{A}) \simeq B^{(0)}(\mathcal{A})$ is a theorem of Costello–Gwilliam. At genus $g \geq 1$, the bar complex acquires curvature $m_0 = \kappa(\mathcal{A}) \cdot \omega_g$ from the Kodaira–Spencer deformation, and the BV complex acquires the BV Laplacian Δ_{BV} as a genus-raising operator.

2.2. **The resolution.** The quartic harmonic discrepancy between the two complexes is

$$\delta_4 = Q^{\mathrm{contact}} \cdot \frac{\kappa}{\mathrm{Im}(\tau)} = Q^{\mathrm{contact}} \cdot m_0.$$

The ratio $\delta_4/m_0 = Q^{\mathrm{contact}} = 10/(c(5c+22))$ is a pure algebraic constant: the $\mathrm{Im}(\tau)$ cancels. In the coderived category $D^{\mathrm{co}}(\mathcal{A})$ of Positselski, the relation $d^2 = m_0 \cdot \mathrm{id}$ implies $m_0 \cdot x = d(d(x)) \in \mathrm{Im}(d)$ for every element x. Therefore δ_4 is exact in D^{co}. Higher-arity discrepancies $\delta_r \propto m_0^{r/2-1}$ are similarly absorbed.

Theorem 2.1 (BV/bar in D^{co})**.** *For every chirally Koszul algebra $\mathcal{A}$, the BV and bar complexes are quasi-isomorphic in $D^{\mathrm{co}}(\mathcal{A})$ at every genus.*

At the chain level, the identification holds unconditionally for shadow-depth classes G, L, C and fails for class M (Virasoro, $\mathcal{W}_N$): the harmonic obstruction $1/\mathrm{Im}(\tau)$ is non-holomorphic and hence not a coboundary of the holomorphic bar differential. The coderived category is the minimal categorical framework in which the identification is universal.

2.3. **The three viewpoints.** The inevitability of the coderived category can be understood from three complementary perspectives.

2.3.1. *The worldsheet perspective.* On a genus-g surface Σ_g, the sewing construction introduces the propagator $P = P_{\mathrm{bar}} + P_{\mathrm{exact}} + P_{\mathrm{harm}}$. The bar complex uses $P_{\mathrm{bar}} = d \log E(z, w)$ (meromorphic, weight 1); the BV complex uses the full P. The discrepancy at arity r is $\delta_r = c_r(\mathcal{A}) \cdot P_{\mathrm{harm}}^{\lfloor r/2 \rfloor - 1}$. The crucial identity: $P_{\mathrm{harm}} = m_0/\kappa$, so $\delta_4 = (Q^{\mathrm{contact}}/\kappa) \cdot m_0$. The discrepancy is proportional to the curvature. In D^{co}, where $m_0 \cdot x = d(d(x))$ is exact, the harmonic modes are cohomologically trivial: they do not contribute to the physical amplitudes.

2.3.2. *The holographic perspective.* In the Costello–Li framework, the bulk theory on AdS_{d+1} uses Witten diagrams with the bulk-to-boundary propagator; the boundary theory uses the bar complex with the OPE. At one loop, the mismatch is the holographic anomaly: the Weyl anomaly coefficient becomes $\delta_4 = Q^{\mathrm{contact}} \cdot m_0$ in the algebraic framework. The coderived resolution says: this anomaly is *exact* in the correct cohomological sense. Holography is exact at the level of cohomology, not representatives.

2.3.3. *The categorical perspective.* $D^{\mathrm{co}}(\mathcal{A})$ is the homotopy category of dg comodules over $B(\mathcal{A})$ with $d^2 = m_0 \cdot \mathrm{id}$ permitted. The bar complex $B(\mathcal{A})$ satisfies $D_B^2 = 0$ always (curvature absorbed into the total differential). The BV complex satisfies $d_{\mathrm{BV}}^2 = \kappa \cdot \omega_g$ — the same curved relation. The coderived quasi-isomorphism identifies the two in the *minimal* category where both are well-defined, without auxiliary strictification choices.

2.4. **The class-by-class mechanism.** The chain-level analysis reveals a hierarchy: class G (no vertices: $\delta_r = 0$), L (Jacobi kills harmonic coupling), C (role separation: fundamental generators $\neq$ quartic source), M (triple role of T prevents decoupling). For M, the stress tensor simultaneously generates the bar complex, sources Q^{contact}, and is contracted by Δ_{BV}: no algebraic identity can kill the coupling. The coderived resolution works at a different level: not by killing δ_4 but by recognising it as $Q^{\mathrm{contact}} \cdot m_0$, a curvature multiple that is exact in D^{co}.

2.5. **Physical significance.** The coderived resolution completes the algebraic foundation for computing string amplitudes from the bar complex: BV quantisation and bar construction produce quasi-isomorphic complexes in D^{co}. The curvature absorption is the algebraic incarnation of the principle that UV divergences on higher-genus worldsheets are artifacts of the perturbative framework. The theorem also resolves the conceptual puzzle of why the scalar genus expansion $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (algebraic, from the bar complex) agrees with the BV effective action (analytic, from the path integral) at every genus: both are representatives of the same D^{co}-cohomology class.

3. Non-principal DS-KD commutation: the (3,2) partition

3.1. **Context.** Drinfeld–Sokolov reduction $\mathrm{DS}_f\colon V_k(\mathfrak{g}) \to W_k(\mathfrak{g}, f)$ for nilpotent $f \in \mathfrak{g}$ produces W-algebras from affine Kac–Moody algebras. Koszul duality $\mathcal{A} \mapsto \mathcal{A}^!$ produces the dual algebra. The fundamental question: does DS commute with Koszul duality?

3.2. **What is proved.**

(i) Principal nilpotent f_{princ} (all types): proved.
(ii) Hook-type partitions $[N-1, 1]$ in type A: proved (Fehily–Creutzig–Linshaw–Nakatsuka–Sato, transport propagation along the Hasse diagram).
(iii) Self-transpose rectangular $(2, 2)$ in $\mathfrak{sl}_4$: proved by self-duality ($c + c' = 14$, self-dual at $c^* = 7$).

3.3. **The open case.**

Problem 3.1. *Let $f = f_{(3,2)}$ in $\mathfrak{sl}_5$. The partition $(3, 2)$ is non-self-transpose (its transpose is $(2, 2, 1)$) and non-hook. The BRST complex has non-abelian $\mathfrak{n}_+$ of dimension 4 in two grades. Does $\mathrm{DS}_{f_{(3,2)}}(V_k(\mathfrak{sl}_5)^!) \simeq \mathrm{DS}_{f_{(3,2)}}(V_k(\mathfrak{sl}_5))^!$?*

The computation requires a $\sim 3000 \times 3000$ BRST matrix. This is the first non-abelian test of DS-KD commutation.

3.4. **Physical motivation.** Non-principal W-algebras arise as boundary algebras of 3d $\mathcal{N} = 4$ gauge theories with non-maximal Nahm pole boundary conditions (Gaiotto–Witten). DS-KD commutation establishes that Koszul duality of the boundary algebra is compatible with varying the Nahm pole type: a form of "duality of boundary conditions" in holography.

4. The genus-5 cross-channel correction

4.1. **The universal N-formula.** For the principal W-algebra $\mathcal{W}_N$ with generators at conformal weights $2, 3, \ldots, N$, the genus-g free energy decomposes as

$$F_g(\mathcal{W}_N, c) = \kappa(\mathcal{W}_N) \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}(\mathcal{W}_N, c),$$

where the cross-channel correction $\delta F_g^{\mathrm{cross}}$ is a universal polynomial in N at each genus, vanishing at $N = 2$ (uniform weight). At genus 2: $\delta F_2 = B_2(N) + A_2(N)/c$ with $A_2(N) = (N-2)(3N^3+14N^2+22N+33)/24$. At genus 3: four terms with polynomial degrees $(1, 3, 5, 7)$ in N. At genus 4: five terms, degree $d_4 = 4$.

Problem 4.1. *Compute $\delta F_5^{\mathrm{cross}}(\mathcal{W}_N, c)$ and extract the six-term universal N-formula. Verify the predicted degree pattern $d_5 = 5$, $n_5 = 6$ terms, net degree 1. Determine $A_{\mathrm{cross}}/A_{\mathrm{scalar}}$ uniquely.*

This requires enumerating all ∼4000 stable graphs of $\overline{\mathcal{M}}_{5,0}$ and computing the gravitational mixed-channel amplitude for each.

4.2. **Physical significance.** The cross-channel correction quantifies the contribution of mixed-propagator graphs: worldsheet configurations where different OPE channels interact. These have no analogue in the scalar (single-channel) theory and represent genuinely new stringy physics. The ratio $A_{\text{cross}}/A_{\text{scalar}}$ measures how much "heavier" the cross-channel instantons are compared to the scalar ones: a universal quantity characterising the non-perturbative structure of multi-weight chiral algebras.

5. Admissible Koszulness at rank ≥ 2

5.1. The rank obstruction.

Conjecture 5.1 (Admissible Koszul rank obstruction)**.** *For every simple Lie algebra $\mathfrak{g}$ of rank $r \geq 2$ and every admissible level $k = p/q - h^\vee$ with denominator $q \geq 3$,* $\dim H^2(B(L_k(\mathfrak{g}))) = \operatorname{rank}(\mathfrak{g})$.

The obstruction arises from the abelian Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$: the Poisson differential d_1 on the Li–bar E_1 page kills all root generators (via $[e_\alpha, f_\alpha] = h_\alpha$) but the Cartan generators $h_1, \ldots, h_r$ survive because $\mathfrak{h}$ is abelian.

At rank 1 ($\mathfrak{sl}_2$): the Cartan is 1-dimensional, but the PBW spectral sequence has at most 2 non-zero columns ($\dim \mathfrak{sl}_2 = 3$), forcing all higher differentials to vanish. Hence $L_k(\mathfrak{sl}_2)$ is Koszul at all admissible levels.

At rank 2 ($\mathfrak{sl}_3$): proved. $H^2 = 2$ at weight $d = q$. At rank 3 ($\mathfrak{sl}_4$): predicted $H^2 = 3$. Computation feasible.

5.2. **Physical significance.** Admissible-level simple quotients are the VOAs of rational conformal field theories (Arakawa). Their non-Koszulness means the bar-cobar inversion fails: the null vectors that make the theory rational simultaneously obstruct PBW concentration. This is the algebraic shadow of the representation-theoretic rigidity of rational CFTs.

6. DK-4 and DK-5: Extending the Drinfeld–Kohno bridge

6.1. **DK-4: Beyond the evaluation core.** MC3 (proved for all simple types) establishes thick generation of the DK category by evaluation modules. DK-4 asks whether this extends to the full module category.

Problem 6.1 (DK-4)**.** *Is the evaluation-generated core of the DK category dense in the full module category of $U_q(\mathfrak{g})$, in the sense that the localisation functor is fully faithful on ind-completions?*

The gap is algebraic: pointwise Ext computations (at individual modules) do not automatically globalise to the category level. For $\mathfrak{sl}_3$, genuinely non-evaluation modules (Whittaker, Wakimoto) exist and must be incorporated.

6.2. DK-5: Categorical E_1 primacy.

Problem 6.2 (DK-5)**.** *Does the ordered factorisation structure on* $\operatorname{Ran}(X)$ *induce a braided monoidal equivalence between the ordered bar category and the representation category of the Yangian $Y(\mathfrak{g})$, not just on the evaluation core but on the full factorisation category?*

This is the categorical incarnation of E_1 primacy: the Yangian and its braided monoidal structure are native to the ordered bar complex, and the passage to the full Ran-space category requires the Francis–Gaitsgory pro-nilpotent completion.

7. Grand completion

Problem 7.1. *Construct a Quillen equivalence (or ∞-categorical equivalence) between the category of chiral algebras on X and the category of conilpotent chiral coalgebras on X, with bar-cobar as the Quillen pair.*

This requires: (a) model category structures on both sides, (b) a cumulant recognition theorem (characterising the image of the bar functor among all coalgebras), and (c) a jet principle (extending formal-disk data to global curve data within the model structure). This is the hardest problem in the programme.

8. Analytic realisation: three-layer gap

Three frameworks address the non-perturbative completion of the formal genus expansion:

(a) The sewing envelope $\mathcal{A}^{\mathrm{sew}}$: Hausdorff completion for sewing-amplitude seminorms. Proved convergent for the standard landscape (polynomial OPE growth + subexponential sector growth).
(b) Conformally flat factorisation homology in IndHilb (Moriwaki 2026): analytic, 1-categorical, metric-dependent.
(c) Coderived shadow invariants $Q_g^{\mathrm{an}}(\mathcal{A})$: categorical, using Positselski's coderived category with curvature $d^2 = m_0 \cdot \mathrm{id}$.

Problem 8.1. *Unify these three frameworks into a single "analytic bar complex" $B^{\mathrm{an}}(\mathcal{A})$ that simultaneously carries the algebraic bar differential, the analytic sewing data, and the coderived curvature.*

9. E_1 Verdier duality on ordered configurations

The Verdier intertwining $\mathbb{D}_{\mathrm{Ran}}(\overline{B}(\mathcal{A})) \simeq \overline{B}(\mathcal{A}^!)$ operates on the symmetric bar over unordered $\mathrm{Ran}(X)$. The ordered bar $\overline{B}^{\mathrm{ord}}(\mathcal{A})$ lives on $\mathrm{Conf}_n^<(X)$, which is *not* a factorisation space. Therefore $\mathbb{D}_{\mathrm{Ran}}(\overline{B}^{\mathrm{ord}})$ is undefined.

Problem 9.1. *Develop a "ribbon* Ran*" framework in which Verdier duality extends to ordered configurations, and establish the ordered Verdier intertwining $\mathbb{D}_{\mathrm{rib}}(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \simeq \overline{B}^{\mathrm{ord}}((\mathcal{A}^!)^{\mathrm{op}})$.*

The correct E_1 analogue uses opposite-coalgebra duality (reversal involution), not Verdier duality. The ribbon Ran development would provide the geometric home for the ordered Koszul dual.

10. Resurgence: the cross-channel instanton action

The scalar instanton action $A_{\mathrm{scalar}} = (2\pi)^2$ is universal (proved). The cross-channel correction $\delta F_g^{\mathrm{cross}}$ has its own instanton action $A_{\mathrm{cross}} > A_{\mathrm{scalar}}$.

Problem 10.1. *Determine $A_{\mathrm{cross}}/A_{\mathrm{scalar}}$ exactly. Current bounds:* $[1.7, 3.1]$. *Genus-5 data would pin this down uniquely.*

The cross-channel instanton is a non-perturbative saddle-point configuration in which different OPE channels interact coherently across the worldsheet.

11. Cross-channel generating function

Problem 11.1. *Does the cross-channel genus expansion $\sum_{g\geq 2} \delta F_g^{\mathrm{cross}}(\mathcal{W}_N, c)\, \hbar^{2g}$ admit a closed-form generating function in the bivariate (g, N)?*

No $\hat{A}$-type generating function exists. The bivariate generating function is non-separable: the N-dependence at genus g involves a polynomial of degree $2g+1$ from power sums of generator weights. The correct framework is the $\mathcal{W}_N$ Frobenius manifold with $N-1$ flat coordinates, not a classical integrable hierarchy.

12. Scalar saturation residual

Conjecture 12.1 (Scalar saturation)**.** *For every modular Koszul algebra $\mathcal{A}$ (including non-algebraic families),* $\dim H^2_{\mathrm{cyc}}(\mathcal{A}) = 1$*: the cyclic deformation complex has a one-dimensional H^2 at every relevant weight.*

A violation would produce "isomodular" algebras: chiral algebras with identical κ but different higher-genus amplitudes. No counterexample is known among 61+ tested families across five independent verification paths.

Each of these problems is a natural outgrowth of the modular Koszul duality programme. Their resolution would extend the reach of the bar complex from the current "standard landscape" to the full world of chiral algebras, from perturbative to non-perturbative, from scalar to multi-channel, from symmetric to ordered, and from algebraic to analytic.